70B check: Qwen/Qwen2.5-72B-Instruct on vLLM 0.32.0 (TP=2, 2x H200)

Qwen/Qwen2.5-72B-Instruct · 2x H200 141 GB (RunPod) · one release

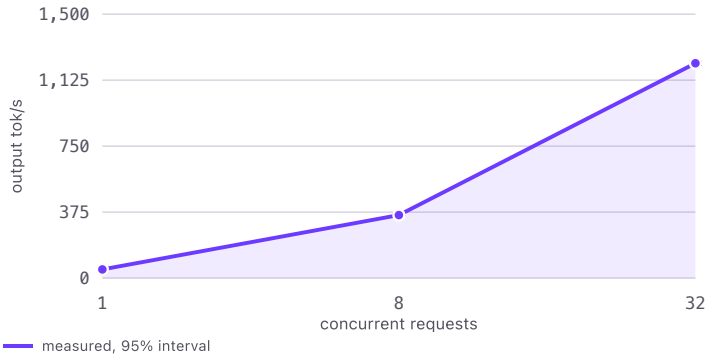
QUALIFIED

DECISION: PROMOTE

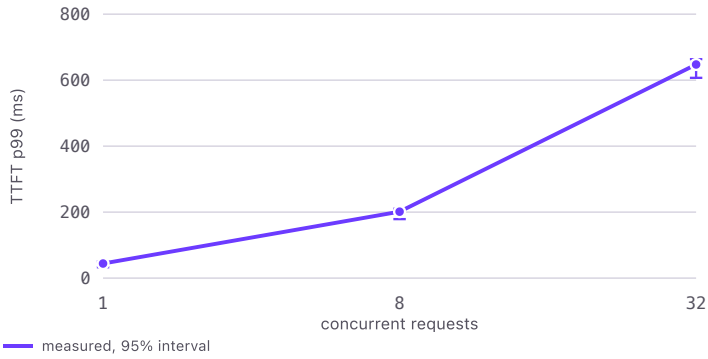
PROMOTE: every gate passed on valid data: the candidate can replace the baseline. **0 failed, 0 inconclusive, 22 passed, 12 skipped** of 34 gates. every gate passed. Exit code 0.

GATE	LIMIT	VALUE	STATUS
ACCURACY 1 pass, 3 skip			
`accuracy.top1_agreement` Top-1 agreement with the reference (teacher-forced)	≥ 98%	**99.02%**	PASS
`accuracy.greedy_exact_match` Greedy output matches the reference (text method)	≥ 70%	—	SKIP
`relative.top1_agreement` Top-1 agreement not more than 0.2 points below the baseline	≥ -0.2 pts vs baseline	—	SKIP
`relative.greedy_exact_match` Greedy match not more than 5 points below the baseline (text method, paired 95% interval)	≥ -5.0 pts vs baseline	—	SKIP
FUNCTIONAL 9 pass			
9 gates: greedy_determinism, max_tokens, stop_sequence, stream_consistency, usage, chat_template, bad_requests, n_choices, seed_reproducible			PASS
STRUCTURED OUTPUT AND TOOLS 2 pass, 2 skip			
`structured.json_schema_valid` JSON output valid against its schema (best supported mode, Wilson 95% interval)	≥ 90%	**100.00%** 95%: 92.87% – 100.00%	PASS
`structured.tool_calls_valid` Tool calls correct (right tool or none, arguments valid against the schema, expected values; Wilson 95% interval)	≥ 90%	**98.92%** 95%: 94.16% – 99.81%	PASS
`relative.json_schema_valid` JSON schema-valid rate not more than 5 points below the baseline	≥ -5.0 pts vs baseline	—	SKIP
`relative.tool_calls_valid` Tool-call valid rate not more than 5 points below the baseline	≥ -5.0 pts vs baseline	—	SKIP
PERFORMANCE 0 pass, 3 skip			
`relative.throughput@1` Output tok/s at least 97% of the baseline (c=1)	≥ 97% of baseline	—	SKIP
`relative.throughput@8` Output tok/s at least 97% of the baseline (c=8)	≥ 97% of baseline	—	SKIP
`relative.throughput@32` Output tok/s at least 97% of the baseline (c=32)	≥ 97% of baseline	—	SKIP
STABILITY 1 pass			
`stability.drift` Throughput drift, start vs end of the stability run	within ±10%	**+6.3%**	PASS
RELIABILITY 8 pass			
8 gates: overload_survives, overload_clean_errors, overload_recovery, cancel_stream_drain, cancel_stream_serves, abandon_drain, abandon_serves, restart_recovery			PASS
REQUESTS AND ERRORS 1 pass			
`errors.error_rate` No failed requests (all checks)	= 0%	**0.00%**	PASS

Output tokens per second



Time to first token, slowest 1% (p99)



70B check: Qwen/Qwen2.5-72B-Instruct on vLLM 0.32.0 (TP=2, 2x H200)

Qwen/Qwen2.5-72B-Instruct on 2x H200 141 GB (RunPod). One release checked on its own; 3 performance repeats; accuracy against a float32 reference, from `qwen2.5-72b-instruct.json`.

RELEASE

vLLM 0.32.0 TP=2

vllm 0.32.0 · 2x H200 141 GB (RunPod) · `http://<host>:18001`

QUALIFIED

DECISION: PROMOTE

PROMOTE: every gate passed on valid data: the candidate can replace the baseline. **0 failed, 0 inconclusive, 22 passed, 12 skipped** of 34 gates. every gate passed. Exit code 0.

TOP-1 AGREEMENT	GREEDY EXACT MATCH	PEAK TOK/S (C=32)	TTFT P50 / TPOT P50 AT C=1	TOOL CALLS VALID
99.02%	**71.9%**	**1,221**	**36 / 20.1 ms**	**98.9%**
teacher-forced, vs reference	logprobs method			93 tasks; JSON schema-valid 100%

STABILITY DRIFT
+6.3%
error rate 0.00%

1. Gates

default gates; relative gates compare the candidate with the baseline and decide on the 95% interval

GATE	TYPE	LIMIT	VALUE	STATUS
ACCURACY 1 pass, 3 skip				
`accuracy.top1_agreement` Top-1 agreement with the reference (teacher-forced)	absolute	≥ 98%	**99.02%**	PASS
`accuracy.greedy_exact_match` Greedy output matches the reference (text method) not applicable: accuracy.method is 'logprobs', not 'text'	absolute	≥ 70%	—	SKIP
`relative.top1_agreement` Top-1 agreement not more than 0.2 points below the baseline no baseline in this run	relative (delta)	≥ -0.2 pts vs baseline	—	SKIP
`relative.greedy_exact_match` Greedy match not more than 5 points below the baseline (text method, paired 95% interval) no baseline in this run	relative (delta)	≥ -5.0 pts vs baseline	—	SKIP
LONG CONTEXT 0 pass, 4 skip				
`long_context.needle_found` Long context *: the answer contains the needle the long_context check did not run	absolute	≥ 80%	—	SKIP
`long_context.top1_agreement` Long context *: top-1 agreement with the reference (teacher-forced) the long_context check did not run	absolute	≥ 95%	—	SKIP
`relative.long_context.needle_found` Long context *: needle found at most one prompt in 5 less often than the baseline the long_context check did not run	relative (delta)	≥ -20.0 pts vs baseline	—	SKIP

GATE	TYPE	LIMIT	VALUE	STATUS
`relative.long_context.top1_agreement` Long context *: top-1 agreement not more than 2 points below the baseline the long_context check did not run	relative (delta)	≥ -2.0 pts vs baseline	—	SKIP
FUNCTIONAL 9 pass				
`functional.greedy_determinism` Functional: greedy_determinism	absolute	= pass	**pass**	PASS
`functional.max_tokens` Functional: max_tokens	absolute	= pass	**pass**	PASS
`functional.stop_sequence` Functional: stop_sequence	absolute	= pass	**pass**	PASS
`functional.stream_consistency` Functional: stream_consistency	absolute	= pass	**pass**	PASS
`functional.usage` Functional: usage	absolute	= pass	**pass**	PASS
`functional.chat_template` Functional: chat_template	absolute	= pass	**pass**	PASS
`functional.bad_requests` Functional: bad_requests	absolute	= pass	**pass**	PASS
`functional.n_choices` Functional: n_choices	absolute	= pass	**pass**	PASS
`functional.seed_reproducible` Functional: seed_reproducible	absolute	= pass	**pass**	PASS
STRUCTURED OUTPUT AND TOOLS 2 pass, 2 skip				
`structured.json_schema_valid` JSON output valid against its schema (best supported mode, Wilson 95% interval)	absolute	≥ 90%	**100.00%** 95%: 92.87% – 100.00%	PASS
`structured.tool_calls_valid` Tool calls correct (right tool or none, arguments valid against the schema, expected values; Wilson 95% interval)	absolute	≥ 90%	**98.92%** 95%: 94.16% – 99.81%	PASS
`relative.json_schema_valid` JSON schema-valid rate not more than 5 points below the baseline no baseline in this run	relative (delta)	≥ -5.0 pts vs baseline	—	SKIP
`relative.tool_calls_valid` Tool-call valid rate not more than 5 points below the baseline no baseline in this run	relative (delta)	≥ -5.0 pts vs baseline	—	SKIP
PERFORMANCE 0 pass, 3 skip				
`relative.throughput@1` Output tok/s at least 97% of the baseline (c=1) no baseline in this run	relative (ratio)	≥ 97% of baseline	—	SKIP
`relative.throughput@8` Output tok/s at least 97% of the baseline (c=8) no baseline in this run	relative (ratio)	≥ 97% of baseline	—	SKIP
`relative.throughput@32` Output tok/s at least 97% of the baseline (c=32) no baseline in this run	relative (ratio)	≥ 97% of baseline	—	SKIP
STABILITY 1 pass				
`stability.drift` Throughput drift, start vs end of the stability run	absolute	within ±10%	**+6.3%**	PASS
RELIABILITY 8 pass				
`reliability.overload_survives` Reliability: overload_survives	absolute	= pass	**pass**	PASS
`reliability.overload_clean_errors` Reliability: overload_clean_errors	absolute	= pass	**pass**	PASS
`reliability.overload_recovery` Reliability: overload_recovery	absolute	= pass	**pass**	PASS

GATE	TYPE	LIMIT	VALUE	STATUS
`reliability.cancel_stream_drain` Reliability: cancel_stream_drain	absolute	= pass	**pass**	PASS
`reliability.cancel_stream_serves` Reliability: cancel_stream_serves	absolute	= pass	**pass**	PASS
`reliability.abandon_drain` Reliability: abandon_drain	absolute	= pass	**pass**	PASS
`reliability.abandon_serves` Reliability: abandon_serves	absolute	= pass	**pass**	PASS
`reliability.restart_recovery` Reliability: restart_recovery	absolute	= pass	**pass**	PASS
REQUESTS AND ERRORS 1 pass				
`errors.error_rate` No failed requests (all checks)	absolute	= 0%	**0.00%**	PASS

2. What was tested

observed = read from the server · declared = from the config

FIELD	RELEASE	CHECK
`engine`	vllm OBSERVED	
`engine_version`	0.32.0 OBSERVED	
`served_model`	Qwen/Qwen2.5-72B-Instruct OBSERVED	
`model_root`	495f39366efef23836d0cfae4fbe635880d2be31 OBSERVED	
`max_model_len`	8192 OBSERVED	
`server_header`	uvicorn OBSERVED	
`hardware`	2x H200 141 GB (RunPod) DECLARED	

Reference: HF transformers 5.20.1, float32, Qwen/Qwen2.5-72B-Instruct revision `495f39366efe`, sha256 `ca20f25146f80080`.

Execution: single

3 performance repeats, 3 slots.

3. Accuracy and functional behaviour

24 prompts vs the reference

LOGPROBS The server returned next-token log-probabilities: every position of the reference's own continuation (1,486 positions in 24 prompts) is scored under the reference's context (teacher forcing), so one early difference does not cascade.

Method	LOGPROBS echo, tokens matched by id; prompts as tokens
Prompts / positions	24 / 1,486
Top-1 agreement	**99.02%** (6 disagreements, 3 at near ties under 0.1 nats); |logprob error| mean 0.0103, max 0.241
Exact ties	17 positions where the server's top log-probability was shared by two or more tokens (17 of them include the reference token). Ties are split: a tie of k tokens that includes the reference token counts 1/k. Ties counted as misses (strict): **98.45%**; ties counted as matches (generous): **99.60%**.
Greedy exact match	**71.9%** mean; 14 of 24 prompts identical to the reference

Lowest-scoring prompts (release)

PROMPT	TOP-1	GREEDY MATCH	FIRST DIFFERENCE AT	REFERENCE VS OUTPUT (START)
`p00` Write a short poem about the sea.	96.1%	15.6%	10	ref: Upon the vast and endless blue, The sea stretches wide, a canvas true. Waves whisper secrets to the shore, In ebb and flow, forever more. Sunset paints the hor out: Upon the vast and endless blue, The sea stretches, a canvas true, Waves whisper secrets to the shore, In ebb and flow, forever more. Sunset paints the horizon'
`p22` Write a haiku about autumn, then explain the syllable structure.	96.1%	53.1%	34	ref: Leaves whisper, falling, Crisp air holds secrets of smoke, Sunset paints gold paths. **Syllable Structure Explanation:** A haiku is a form of poetry that tradi out: Leaves whisper, falling, Crisp air holds secrets of smoke, Sunset paints gold paths. **Syllable Structure Explanation:** A haiku is a traditional form of Japan
`p17` Give a short history of the printing press.	97.7%	1.6%	1	ref: The history of the printing press is a significant chapter in the development of human communication and technology. Here's a brief overview: ### Early Printin out: The printing press is one of the most significant inventions in human history, revolutionizing the way information is disseminated and preserved. Here's a brief
`p03` Give me a recipe for a vegetarian lasagna.	97.7%	18.8%	12	ref: Certainly! Here's a delicious vegetarian lasagna recipe that's sure to please. This recipe serves 6-8 people. ### Ingredients: #### For the Tomato Sauce: - 2 out: Certainly! Here's a delicious vegetarian lasagna recipe that's both comforting and packed with flavor. This recipe serves 6-8 people. ### Ingredients: #### Fo
`p06` A train leaves at 3 pm going 60 km/h; another at 4 pm going 90 km/h. When does the second catch up? Show the steps.	97.7%	42.2%	27	ref: To determine when the second train catches up to the first, we need to set up an equation based on the distances traveled by both trains. ### Step 1: Define th out: To determine when the second train catches up to the first, we need to set up an equation based on the distances traveled by both trains. Let's break it down st

Every prompt is in accuracy.csv.

Functional checks

CHECK	RESULT	DETAIL
Greedy determinism `greedy_determinism`	PASS	two identical requests gave the same 209-character output
max_tokens honoured `max_tokens`	PASS	finish_reason length, usage.completion_tokens 8
Stop sequence honoured `stop_sequence`	PASS	output ended before 'ets ' with finish_reason stop
Streaming equals non-streaming `stream_consistency`	PASS	identical text
Usage counts present and consistent `usage`	PASS	prompt 37, completion 16, total 53; streamed usage matches
Chat template matches the reference's `chat_template`	PASS	chat output equals completions on the templated ids; chat prompt_tokens 37 = 37
Bad requests get a 4xx `bad_requests`	PASS	malformed JSON: 400; unknown model: 404; negative max_tokens: 400; token id out of vocabulary: 400
n=2 returns two choices `n_choices`	PASS	two choices
Same seed, same sampled output `seed_reproducible`	PASS	identical sampled outputs

Reliability overload, cancelled and abandoned requests, restart

CHECK	RESULT	DETAIL
Survives an overload burst `overload_survives`	PASS	the server answers /v1/models after the burst
Overload: requests complete or are refused cleanly (429 / 503) `overload_clean_errors`	PASS	256 completed, 0 refused cleanly (429 / 503), 0 other errors
Back to idle latency after the overload `overload_recovery`	PASS	a short request took 337 ms idle and was back within 3x 0.34 s after the burst
Cancelled streams stop running on the server `cancel_stream_drain`	PASS	16 streams cut off; running requests back to 0 after 0.66 s
Serves normally after cancelled streams `cancel_stream_serves`	PASS	a short request completed in 336 ms
Abandoned non-streaming requests stop running on the server `abandon_drain`	PASS	16 requests cut off; running requests back to 0 after 0.04 s
Serves normally after abandoned requests `abandon_serves`	PASS	a short request completed in 336 ms
Recovers from a restart with the same output `restart_recovery`	PASS	back after 70.4 s with the same greedy output

4. Structured output and tool calls tasks tools-v1 (101) and json-v1 (50); greedy; the gates decide on the Wilson 95% interval

CHECK	RESULT	NOTE
JSON, json schema gated	**100%** valid (50/50, [92.9%, 100.0%])	parse 100%, expected values 98%; wrong_value 1
JSON, json object	**100%** valid (50/50, [92.9%, 100.0%])	parse 100%, expected values 98%; wrong_value 1
JSON, instruction (instruction only)	**100%** valid (50/50, [92.9%, 100.0%])	parse 100%, expected values 98%; wrong_value 1
Tool calls, tool_choice auto gated	**99%** valid (92/93, [94.2%, 99.8%])	parallel 8/8 (not gated); no_tool_call 1
Tool calls, tool_choice named	**100%** valid (81/81, [95.5%, 100.0%])	
Tool calls, tool_choice named strict	**96%** valid (78/81, [89.7%, 98.7%])	wrong_value 3

Tool calls by category valid / tasks per tool_choice mode; parallel is reported, not gated

CATEGORY	TASKS	AUTO	NAMED	NAMED STRICT	ERRORS (AUTO)
string enum	9	9/9	9/9	9/9	
integer number	9	9/9	9/9	9/9	
boolean	8	8/8	8/8	8/8	
nested object	8	8/8	8/8	8/8	
array	8	8/8	8/8	8/8	
optional args	8	7/8	8/8	7/8	no_tool_call 1
required args	9	9/9	9/9	9/9	
tool selection	10	10/10	10/10	10/10	
parallel	8	8/8	–	–	
irrelevance	8	8/8	–	–	
multi turn	8	8/8	4/4	4/4	

CATEGORY	TASKS	AUTO	NAMED	NAMED STRICT	ERRORS (AUTO)
long schema	8	8/8	8/8	6/8	

JSON by category schema-valid / tasks per mode

CATEGORY	TASKS	JSON SCHEMA	JSON OBJECT	INSTRUCTION	ERRORS (JSON SCHEMA)
flat	7	7/7	7/7	7/7	
enum	7	7/7	7/7	7/7	
nested	7	7/7	7/7	7/7	
array	8	8/8	8/8	8/8	
nullable	7	7/7	7/7	7/7	
numbers	7	7/7	7/7	7/7	
long schema	7	7/7	7/7	7/7	wrong_value 1

Tool-call failures by type auto; first tasks and argument paths

TYPE	COUNT	TASKS	PATHS
`no_tool_call`	1	op.contact	

Evidence request / response samples of failed checks (truncated)

Evidence · json.json_schema · candidate `candidate:json.json_schema:0`

wrong_value: $.address.street: "Keizersgracht" does not contain 'keizersgracht 120'

request: {"task": "j.listing", "mode": "json_schema", "content": "Extract the listing as JSON: 'Bright 3-bedroom apartment, 2 bathrooms, 96 m2 on the 4th floor at Keizersgracht 120, Amsterdam. Built in 1931, balcony, no parking, pets allowed. Available from 2026-12-01 for 2450 EUR a month.'", "schema": {"type": "object", "properties": {"type": {"type": "string", "enum": ["apartment", "house", "studio"]}, "bedrooms": {"type": "integer"}, "bathrooms": {"type": "number"}, "area_m2": {"type": "number"}, "floor": {"type": ["integer", "null"]}, "address": {"type": "object", "properties": {"street": {"type": ... [716 more chars]
response: {"message": {"role": "assistant", "content": "{\n \"type\": \"apartment\",\n \"bedrooms\": 3,\n \"bathrooms\": 2,\n \"area_m2\": 96,\n \"floor\": 4,\n \"address\": {\n \"street\": \"Keizersgracht\",\n \"city\": \"Amsterdam\"\n },\n \"year_built\": 1931,\n \"balcony\": true,\n \"parking\": false,\n \"pets_allowed\": true,\n \"available_from\": \"2026-12-01\",\n \"rent\": {\n \"amount\": 2450,\n \"currency\": \"EUR\",\n \"period\": \"month\"\n }\n}"}, "finish_reason": "stop", "usage": {"prompt_tokens": 115, "total_tokens": 267, "completion_tokens": 152, "prompt_tokens_details": null, "completion_tokens_details": null}}

Evidence · json.json_object · candidate `candidate:json.json_object:1`

wrong_value: $.autoscaling.target_cpu_percent: 0.7, expected 70

request: {"task": "j.config", "mode": "json_object", "content": "Return the service config as JSON: 'Service billing-api, 3 replicas, 2 CPU and 4096 MB memory each, port 8080, health check path /healthz every 15 seconds, log level warn, autoscaling on between 3 and 10 replicas at 70% CPU.' Reply with only the JSON object. It must match this JSON Schema: {\"type\": \"object\", \"properties\": {\"service\": {\"type\": \"string\"}, \"replicas\": {\"type\": \"integer\"}, \"resources\": {\"type\": \"object\", \"properties\": {\"cpu\": {\"type\": \"number\"}, \"memory_mb\": {\"type\": \"integer\"}}, \"requir... [1925 more chars]
response: {"message": {"role": "assistant", "content": "{\n \"service\": \"billing-api\",\n \"replicas\": 3,\n \"resources\": {\n \"cpu\": 2,\n \"memory_mb\": 4096\n },\n \"port\": 8080,\n \"health_check\": {\n \"path\": \"/healthz\",\n \"interval_s\": 15\n },\n \"log_level\": \"warn\",\n \"autoscaling\": {\n \"enabled\": true,\n \"min_replicas\": 3,\n \"max_replicas\": 10,\n \"target_cpu_percent\": 0.7\n }\n}"}, "finish_reason": "stop", "usage": {"prompt_tokens": 390, "total_tokens": 521, "completion_tokens": 131, "prompt_tokens_details": null, "completion_tokens_details": null}}

Evidence · json.instruction · candidate candidate:json.instruction:2

wrong_value: $.autoscaling.target_cpu_percent: 0.7, expected 70

request: {"task": "j.config", "mode": "instruction", "content": "Return the service config as JSON: 'Service billing-api, 3 replicas, 2 CPU and 4096 MB memory each, port 8080, health check path /healthz every 15 seconds, log level warn, autoscaling on between 3 and 10 replicas at 70% CPU.' Reply with only the JSON object. It must match this JSON Schema: {\"type\": \"object\", \"properties\": {\"service\": {\"type\": \"string\"}, \"replicas\": {\"type\": \"integer\"}, \"resources\": {\"type\": \"object\", \"properties\": {\"cpu\": {\"type\": \"number\"}, \"memory_mb\": {\"type\": \"integer\"}}, \"requir... [1925 more chars]
response: {"message": {"role": "assistant", "content": "{\n \"service\": \"billing-api\",\n \"replicas\": 3,\n \"resources\": {\n \"cpu\": 2,\n \"memory_mb\": 4096\n },\n \"port\": 8080,\n \"health_check\": {\n \"path\": \"/healthz\",\n \"interval_s\": 15\n },\n \"log_level\": \"warn\",\n \"autoscaling\": {\n \"enabled\": true,\n \"min_replicas\": 3,\n \"max_replicas\": 10,\n \"target_cpu_percent\": 0.7\n }\n}"}, "finish_reason": "stop", "usage": {"prompt_tokens": 390, "total_tokens": 521, "completion_tokens": 131, "prompt_tokens_details": null, "completion_tokens_details": null}}

Evidence · tools.auto · candidate candidate:tools.auto:3

no_tool_call: no tool_calls in the reply (content: Sure, I can help with that. Could you please provide me with more details such as his email, phone number, and address?)

request: {"task": "op.contact", "mode": "auto", "messages": [{"role": "user", "content": "Add a contact named Tom Becker."}], "tool_choice": "auto", "tools": ["create_contact"], "expected_tool_parameters": {"create_contact": {"type": "object", "properties": {"first_name": {"type": "string"}, "last_name": {"type": "string"}, "email": {"type": "string", "description": "An email address."}, "phone": {"type": "string"}, "address": {"type": "object", "properties": {"street": {"type": "string"}, "city": {"type": "string"}, "postal_code": {"type": "string"}, "country": {"type": "string"}}, "required": ["city"... [120 more chars]
response: {"message": {"role": "assistant", "content": "Sure, I can help with that. Could you please provide me with more details such as his email, phone number, and address?"}, "finish_reason": "stop", "usage": {"prompt_tokens": 282, "total_tokens": 310, "completion_tokens": 28, "prompt_tokens_details": null, "completion_tokens_details": null}}

Evidence · tools.named_strict · candidate candidate:tools.named_strict:4

wrong_value: search_products $.in_stock_only: true was not asked for (invented)

request: {"task": "op.search", "mode": "named_strict", "messages": [{"role": "user", "content": "Search the catalog for running shoes."}], "tool_choice": {"type": "function", "function": {"name": "search_products"}}, "tools": ["search_products"], "expected_tool_parameters": {"search_products": {"type": "object", "properties": {"query": {"type": "string"}, "category": {"type": "string", "enum": ["electronics", "clothing", "home", "sports", "books"]}, "max_price": {"type": "number"}, "in_stock_only": {"type": "boolean", "default": false}, "sort": {"type": "string", "enum": ["relevance", "price_asc", "pri... [102 more chars]
response: {"message": {"role": "assistant", "tool_calls": [{"id": "chatcmpl-tool-b4298ba6be5f81d4", "type": "function", "function": {"name": "search_products", "arguments": "{\"query\": \"running shoes\", \"category\": \"sports\", \"in_stock_only\": true, \"sort\": \"relevance\"}"}}]}, "finish_reason": "stop", "usage": {"prompt_tokens": 266, "total_tokens": 307, "completion_tokens": 41, "prompt_tokens_details": null, "completion_tokens_details": null}}

Evidence · tools.named_strict · candidate candidate:tools.named_strict:5

wrong_value: create_user_account $.timezone: missing, expected "Asia/Kolkata"

request: {"task": "lo.user_contractor", "mode": "named_strict", "messages": [{"role": "user", "content": "Set up a contractor account: Ravi Patel, username rpatel, ravi.patel@corp.example, support department, start 2026-12-01, remote, no laptop, time zone Asia/Kolkata, groups helpdesk and kb-editors."}], "tool_choice": {"type": "function", "function": {"name": "create_user_account"}}, "tools": ["create_user_account", "set_account_active"], "expected_tool_parameters": {"create_user_account": {"type": "object", "properties": {"username": {"type": "string", "description": "Lower-case login name."}, "email... [1192 more chars]
response: {"message": {"role": "assistant", "tool_calls": [{"id": "chatcmpl-tool-a73127abcd28db37", "type": "function", "function": {"name": "create_user_account", "arguments": "{\"username\": \"rpatel\", \"email\": \"ravi.patel@corp.example\", \"full_name\": \"Ravi Patel\", \"role\": \"contractor\", \"department\": \"support\", \"start_date\": \"2026-12-01\", \"office\": \"remote\", \"laptop\": \"none\", \"groups\": [\"helpdesk\", \"kb-editors\"], \"is_contractor\": true, \"send_welcome_email\": true}"}}]}, "finish_reason": "stop", "usage": {"prompt_tokens": 667, "total_tokens": 780, "completion_tokens": 113, "prompt_tokens_details": null, "completion_tokens_details": null}}

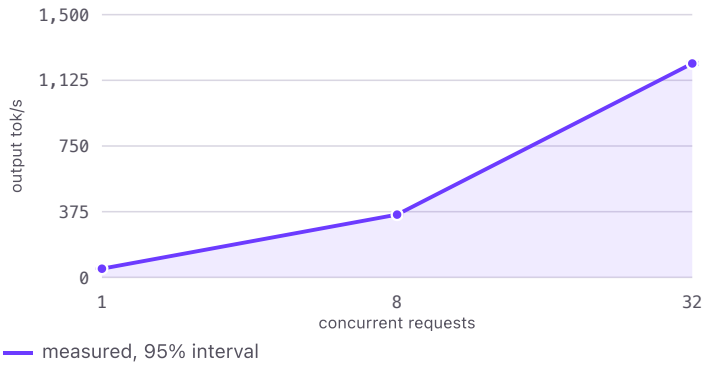
Evidence · tools.named_strict · candidate `candidate:tools.named_strict:6`

wrong_value: create_job_posting $.languages: missing, expected ["English", "French"]
request: {"task": "lo.job_intern", "mode": "named_strict", "messages": [{"role": "user", "content": "Post an onsite internship: Marketing Intern, Marketing department, Toronto, starting 2027-01-11, English and French required."}], "tool_choice": {"type": "function", "function": {"name": "create_job_posting"}}, "tools": ["create_job_posting"], "expected_tool_parameters": {"create_job_posting": {"type": "object", "properties": {"title": {"type": "string"}, "department": {"type": "string"}, "location": {"type": "string"}, "remote": {"type": "string", "enum": ["onsite", "hybrid", "remote"]}, "employment_ty... [754 more chars]
response: {"message": {"role": "assistant", "tool_calls": [{"id": "chatcmpl-tool-8ffb50897ef82fb0", "type": "function", "function": {"name": "create_job_posting", "arguments": "{\"title\": \"Marketing Intern\", \"department\": \"Marketing\", \"location\": \"Toronto\", \"remote\": \"onsite\", \"employment_type\": \"internship\", \"start_date\": \"2027-01-11\", \"description\": \"We are looking for a motivated Marketing Intern to join our team in Toronto. The ideal candidate will have a passion for marketing and a desire to learn and grow within the industry. This is an onsite position, and fluency in English and French is required. The internship will start on January 11, 2027. Please apply if you are excited about this opportunity and meet the requirements.\"}"}}]}, "finish_reason": "stop", "usage": {"prompt_tokens": 452, "total_tokens": 604, "completion_tokens": 152, "prompt_tokens_details": null... [37 more chars]

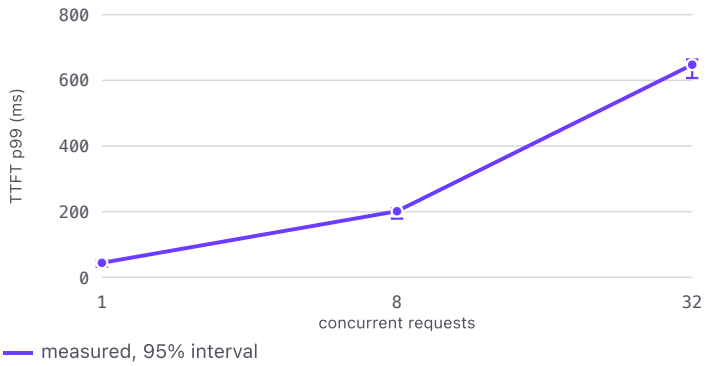
5. Performance

short: 128 in / 128 out per request; 3 repeats per point

Output tokens per second



Time to first token, slowest 1% (p99)



Release: vLLM 0.32.0 TP=2

LOAD	ACHIEVED	REQUESTS	ERRORS	OUTPUT TOK/S (95% CI)	TTFT P50 / P99 MS	TPOT P50 / P99 MS	ITL P99 MS	E2E P50 / P99 MS
c=1	c 1.0	48	0	**49.3** 49 – 49	36 / 44	20.14 / 20.15	20.92	2,594 / 2,602
c=8	c 8.0	48	0	**357.8** 355 – 361	182 / 202	21.06 / 21.72	24.72	2,855 / 2,880
c=32	c 31.8	192	0	**1,221.3** 1,213 – 1,229	596 / 647	21.52 / 24.95	29.51	3,340 / 3,402

TTFT: time to the first streamed token. TPOT: time per output token after the first, (E2E - TTFT) / (tokens - 1), per request. ITL: gaps between streamed chunks. Achieved concurrency: mean requests in flight. Chart whiskers: 95% interval over the repeats. A gate passes or fails only when its whole 95% interval is on one side of the limit; otherwise it is INCONCLUSIVE.

6. Stability and memory

180 s soak at 16 concurrent requests

METRIC	RELEASE
Requests completed	800
Errors	0
Output tok/s overall	558.1
Output tok/s, first 10%	546.1
Output tok/s, last 10%	580.6

METRIC	RELEASE
Throughput drift, first vs last 10%	+6.3%
TTFT p99 first / last 10% (ms)	1,028 / 963
TTFT p99 drift	-6.3%
Memory source	prometheus
KV-cache usage, max	2.1%
KV-cache usage, mean during stability	-
Preemptions	0
API server RSS start / end (MiB)	1,508 / 1,470
RSS growth during stability (MiB)	-

First and last 10% of the run after the ramp (3.7 s, one median request time) is excluded; tokens are counted when they stream.

nvidia-smi on the validator's machine: GPU memory max 262,938 MiB during the soak (nvidia-smi on the validator's machine: every GPU (memory summed), shared by every server on it).

7. Not in this version and not checked in this run

V1 Automatic component-level root cause V1 Signed, independent attestation of results V1 Cross-customer inference intelligence

V2 Automated workload routing and rollback

- **long_context**: not requested (checks:)
- **gate long_context.needle_found**: the long_context check did not run
- **gate long_context.top1_agreement**: the long_context check did not run
- **gate relative.long_context.needle_found**: the long_context check did not run
- **gate relative.long_context.top1_agreement**: the long_context check did not run

Reserved in report.json for later versions: root_cause, attestation, fleet_intelligence, routing.

8. Reproduce

```
python -m fusezap validate --config results/qwen2.5-72b-instruct.json
```

Config qwen2.5-72b-instruct.json sha256 `2a273d9270119de5`; reference sha256 `ca20f25146f80080`; seed 0. Tool 0.1.0 at commit `4c66918`. Exit code 0. Validate runs entirely on your systems and sends nothing to FuseZap.

Client: Python 3.12.15 on Linux-6.8.0-147-generic-x86_64-with-glibc2.35 (validator host).